

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Shortage of {donors/ kidneys}/ (dangers of) { surgery/ long term immunosuppresants}/ may need to be replaced in the future/ may have to wait a long time for a kidney	1			1		
		(ii)		antigens <u>on cells</u> of donor kidneys (1) white blood cells produce antibodies (specific to antigens) (1) that {destroy/ act against} the {antigens/cells} (1) Reject {kill/ fight/ attack} antigens		3		3		
	(b)	(i)		myeloma/ tumour cells/cancerous white cells/ cancer cells	1			1		
		(ii)		lymphocytes	1			1		
		(iii)		hybridoma	1			1		
		(iv)	B: C:	(Injected) antigen {stimulates/ causes} {immune response/ lymphocyte production/ cloning of lymphocytes} (1) {cells B/ lymphocytes} and {cells A/ myeloma} are {fused/ combine/ merge} together (1)		2		2		
	(c)			Any two (x1) from: • diagnosis of [disease/chlamydia/HIV] • tissue typing for transplants • monitoring of the spread of malaria • supporting chemotherapy for cancers • pregnancy testing	2			2		
				Question 4 total	6	5	0	11	0	0